

DECODE THE CHART – DETAILED ANSWERS

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CHART 1 – Regional Comparison

1. What is it trying to tell you?

This bar chart is designed to **compare football sales in four different regions during the second quarter of 2026**. The regions shown are **Indore, Surat, Pune, and Vizag**. The height of each bar represents the amount of football sales in that particular region.

From the chart, **Surat has the highest football sales**, while **Indore has the lowest sales**. Pune and Vizag have sales between these two regions. Therefore, the main purpose of this chart is to help us understand and compare the sales performance of different regions during Q2 2026.

2. Do you trust it?

I trust the chart only partially.

3. Why or why not?

The chart is easy to understand because it uses a simple bar chart and clearly displays four regions. However, there is an important problem with the **vertical axis**. The y-axis does not start from zero; it starts around **440**. Because of this, the differences between the bars can appear much larger than the actual differences in sales.

For example, Surat appears significantly higher than Indore because the chart is using a shortened scale. If the y-axis started from zero, the difference between the regions would appear much smaller. Therefore, the data may be correct, but the **visual presentation can exaggerate the difference**.

Conclusion: The chart can be used for comparison, but the scale of the y-axis should be checked before making a strong conclusion.

CHART 2 – Expense Breakdown by Category

1. What is it trying to tell you?

This pie chart shows the **breakdown of total expenses into eight different categories**. Each section or slice of the pie represents one expense category. The size of each slice indicates how much of the total expenditure belongs to that category.

The **larger slices represent categories where more money is being spent**, while the smaller slices represent categories that contribute less to the total expenses. The purpose of this chart is to give a quick visual understanding of how the total budget or expenses are distributed among different categories.

2. Do you trust it?

I trust it to give a general idea, but I would not use it for precise comparisons.

3. Why or why not?

The chart provides a quick overview of the expense distribution, but there are some limitations. There are **many categories and several similarly sized slices**, which makes it difficult to compare them accurately.

Also, the chart does not clearly show the **exact percentage or monetary value** for every category. Without these numbers, it is difficult to determine the precise difference between two similar-sized sections.

A **bar chart with category names and exact percentages or amounts** would be easier to read and compare. Therefore, the pie chart is useful for understanding the overall distribution, but it is not the best choice for detailed analysis.

Conclusion: I would use this chart for a general overview, but I would prefer a bar chart for accurate comparison.

CHART 3 – Movie Ticket Sales vs. Umbrella Sales

1. What is it trying to tell you?

This line chart compares **movie ticket sales and umbrella sales over different time periods**. It contains two lines representing the two types of sales.

The chart shows that both movie ticket sales and umbrella sales generally **increase over time**, although there are some fluctuations. Because the two lines appear to move in a somewhat similar direction, the chart may suggest that there is a relationship or correlation between movie ticket sales and umbrella sales.

For example, when movie ticket sales increase, umbrella sales sometimes increase as well. However, simply observing two lines moving together does not mean that one factor is directly causing the other.

2. Do you trust it?

I do not completely trust the conclusion that the two variables are directly related.

3. Why or why not?

The chart shows a possible relationship between the two types of sales, but **correlation does not mean causation**. Just because movie ticket sales and umbrella sales increase at similar times does not mean that people buying movie tickets causes them to buy umbrellas.

There could be another factor affecting both sales. For example, **weather conditions, rainy seasons, holidays, weekends, festivals, or special events** could influence both movie attendance and umbrella purchases.

Also, the chart appears to contain only a limited number of time periods. More data collected over a longer period would be needed to determine whether the relationship is genuine.

Conclusion: The chart can show that the two variables may be correlated, but it cannot prove that one causes the other. Additional information and a larger dataset would be required.

CHART 4 – Study Hours vs. Exam Score

1. What is it trying to tell you?

This scatter plot shows the **relationship between the number of hours students study and their exam scores**. The horizontal axis represents **study hours**, while the vertical axis represents the **exam score**.

Each dot represents an individual student's data. By looking at the positions of the dots, we can see whether students who study more hours generally achieve higher examination scores.

The points are widely spread across the chart. There does not appear to be a very strong upward or downward pattern. This suggests that **study hours alone may not have a strong relationship with exam scores in this dataset**.

Students who study a similar number of hours can have very different exam scores. Similarly, some students who study fewer hours may still obtain relatively high scores.

2. Do you trust it?

Yes, I trust the scatter plot as a representation of the data, but I would not conclude that study hours alone determine exam scores.

3. Why or why not?

The chart contains a large number of observations, which makes it useful for identifying the general pattern in the data. However, the points are widely scattered, and there is **no clear strong relationship** between study hours and exam scores.

Exam performance can depend on many other factors, such as:

- Quality of studying
- Previous knowledge
- Concentration
- Study methods
- Class attendance
- Understanding of the subject
- Exam difficulty
- Sleep and rest
- Stress and preparation
- Individual learning ability

Therefore, simply increasing study hours does not automatically guarantee a higher exam score.

A **correlation coefficient or trend line** could be added to the chart to measure the strength and direction of the relationship more accurately.

Conclusion: The chart suggests that there is no strong relationship between study hours and exam scores in this particular dataset. However, this does not mean studying is unimportant; it means that **study hours alone cannot explain exam performance**.